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WORK EXPERIENCE

WEB SOFTWARE ENGINEER INTERN – NOKIA NETWORKS SRL – 02/03/2026 – Current – TIMIȘOARA, ROMANIA

- **Developed and maintained robust backend services** using **Python**, ensuring high-performance data processing for internal web applications.
- **Architected interactive data visualization suites** in **Power BI**, translating complex system metrics into actionable business intelligence reports for stakeholders.
- **Optimized relational database performance** across **MySQL and MariaDB** environments, implementing schema improvements and efficient query logic.
- **Streamlined the software delivery lifecycle** by configuring **Jenkins** automation pipelines, reducing manual deployment time and increasing development velocity.

AI & MACHINE LEARNING TRAINEE – BITDEFENDER – 23/04/2026 – 22/06/2026 – TIMIȘOARA, ROMANIA

Completed an intensive AI/ML program covering Data Science fundamentals (NumPy, Pandas, Matplotlib), Classical ML (Random Forests, KNN), and Deep Learning (CNNs, Keras, TensorFlow).
Capstone Project - InterroAI:

- Built a local desktop multi-agent orchestration platform using **React, Electron, FastAPI, and ChromaDB**.
- Implemented semantic RAG-based indexing with file vector embeddings and offline intent classification (**TF-IDF + Logistic Regression**).
- Engineered an autonomous "Plan -> Code -> Verify" development loop with self-correction via static analysis and test feedback.

FULL-STACK DVELOPER + DEVOPS (FREELANCER) – RUS S ANCA PFA – 16/12/2025 – Current – ARAD, ROMANIA

- **Architected a full-stack e-commerce platform** featuring a responsive React storefront and a secure administrative dashboard.
- **Built a scalable Node.js/TypeScript backend** with Drizzle ORM and Neon PostgreSQL, utilizing Cloudflare R2 for optimized media delivery.
- **Implemented race-condition protections** in the order service to ensure 100% inventory accuracy during peak concurrent traffic.
- **Engineered a robust security layer** featuring RBAC, JWT-based authentication, and Zod validation to mitigate common vulnerabilities.
- **Automated deployment via Docker and Railway**, leveraging Cloudflare's CDN to reduce latency and streamline CI/CD workflows.
- **Integrated Stripe and custom payment workflows**, supporting both digital transactions and pay-on-delivery logic.

FULL-STACK DEVELOPER (FREELANCER) – MASCO-BAITS S.R.L. – 20/05/2025 – 11/08/2025 – ARAD, ROMANIA

- **Developed a high-performance, SEO-optimized e-commerce platform** for competitive fishing equipment using React and Node.js.
- **Engineered a secure MySQL-backed authentication system**, implementing industry-standard account management and data protection.
- **Integrated diverse payment solutions**, including automated Stripe/bank transfer workflows and Pay-on-Delivery logic.
- **Optimized Core Web Vitals** to ensure fast loading speeds and a seamless responsive experience across all device types.
- **Automated deployment and hosting via Railway**, ensuring high availability and a streamlined CI/CD pipeline.

EDUCATION AND TRAINING

29/09/2025 – CURRENT Timișoara, Romania
STUDENT Politehnica University Timișoara

Website <https://ac.upt.ro/> | **Field of study** Automatică și Calculatoare | **Level in EQF** EQF level 6

Website <https://www.tichindeal.ro/> | Level in EQF EQF level 5

SKILLS

Java | JavaScript | Nodejs | HTML | CSS | React | TailwindCSS | C | C# | Python | SQL | NoSQL | Docker | Kubernetes | Git | Linux | Express | PostreSQL | MySQL | MongoDB | C++ | OOP | Microsoft Office (Excel PowerPoint Word) - nivel intermediar

HONOURS AND AWARDS

29/03/2026
1st place - iTEC Timișoara hackathon - web development section – Inspectoratul Școlar Județean Timiș

01/08/2025
Bronze Medal – National Olympiad in Digital Innovation and Creation (Web Development) – Ministerul Educației Naționale

10th place at the National Olympiad

18/04/2025
National Informatics Olympiad – Ministerul Educației Naționale

55th place out of 100

19/04/2025
First Prize – County Informatics Olympiad – Ministerul Educației Naționale

21/04/2024
Second Prize – County Informatics Olympiad – Ministerul Educației Naționale

28/05/2023
Bronze Medal – National Information Technology Olympiad (TIC Section) – Ministerul Educației, Inspectoratul Școlar Județean Dâmbovița

22/04/2023
First Prize – National Information Technology Olympiad (Regional Stage), 2023 – Inspectoratul Școlar Județean Arad

10/05/2023
Third Prize – "Caius Iacob" International Mathematics & Informatics Competition – Universitatea "Aurel Vlaicu"-Arad, ISJ Arad

12/05/2022
Second Prize – National Information Technology Olympiad (Regional Stage), 2022 – ISJ Arad

PROJECTS

iTECify

- iTECify | Lead Developer (iTEC 2026 Web Track – 1st Place Winner)
- **Infrastructure:** Built a high-concurrency monorepo using **Bun Workspaces** and **Turborepo**, deploying a **React 19** frontend and a **Hono/Drizzle** backend.
 - **Security & Isolation:** Designed a multi-stage security pipeline including regex-based static analysis and **Docker-containerized** execution environments with 2GB RAM limits and 3-second timeouts.
 - **Real-Time Systems:** Integrated **Y-Monaco** and **xterm.js** over WebSockets to synchronize code buffers and PTY terminal sessions across global clients in real time.
 - **AI Integration:** Implemented a tool-calling architecture allowing an LLM to interact with the file system and terminal, including a custom "Accept/Reject" diff overlay for AI-suggested changes.
 - **DevOps:** Leveraged **Cloudflare Tunnels** for zero-config local-to-web exposure and integrated the **Vercel API** to provide users with automated, one-click production deployments.
 - **Data Layer:** Engineered a hybrid database strategy using **MongoDB Atlas** for social/user data and **SQLite (Drizzle)** for high-speed file-system indexing and snapshotting.

InterroAI

- Developed an AI-assisted desktop development environment with direct, secure local workspace access.
- **Tech Stack:** React 18, Vite, Electron, Python, FastAPI, WebSockets, ChromaDB, OpenAI API, scikit-learn.
 - **Architecture & Backend:** Built an asynchronous FastAPI backend paired with Electron for native window management, driving real-time agent orchestration via WebSockets.

- **Semantic RAG Pipeline:** Engineered a 2-phase asynchronous project indexing system utilizing language-aware AST chunking and OpenAI embeddings (text-embedding-3-small) stored in a local ChromaDB instance.
- **Offline Hybrid Routing:** Implemented a sub-millisecond, offline intent classifier (**TF-IDF + Logistic Regression**) to dynamically route prompts across model tiers based on task complexity.
- **Autonomous Coding Loop:** Designed a self-correcting **Plan -> Code -> Verify** agent engine that applies programmatic diff patches, executes sandbox testing (pytest/ruff), and performs autonomous error correction.
- **Secure Storage:** Integrated the OS-native keychain (keyring) to ensure hardware-encrypted, at-rest credential management for API keys.

Masco Baits E-Commerce Store

Masco Baits | Full-Stack Developer | React, Node.js, MySQL, Express

- **Engineered** a full-scale e-commerce platform for a specialized retailer, managing the end-to-end product lifecycle from inventory intake to customer fulfillment.
- **Architected** a secure **RBAC (Role-Based Access Control)** system, streamlining administrative workflows for order processing and inventory auditing while securing customer data.
- **Designed** a normalized MySQL database schema to manage high-complexity relationships, including multi-status order tracking, category hierarchies, and user-generated reviews.
- **Boosted** organic discoverability by implementing an SEO strategy with dynamic metadata and semantic HTML, achieving a **95+ Lighthouse performance score**.
- **Integrated** automated transactional email systems and **GDPR-compliant** data handling to ensure operational reliability and regional legal alignment.

Link <https://www.masco-baits.ro/home>

TaskLink

TaskLink | Lead Full-Stack Developer (MERN)

- **Backend:** Built a scalable RESTful API with Node.js and Express, implementing JWT authentication and MongoDB aggregation pipelines for efficient data retrieval.
- **AI Integration:** Utilized Google Cloud Natural Language API to analyze skill sets and automate resume optimization, enhancing user profile SEO.
- **Real-Time Systems:** Implemented a bi-directional WebSocket infrastructure to handle concurrent chat sessions and system-wide alerts.
- **Frontend:** Designed a responsive React.js dashboard featuring a dual-portal architecture and a custom-built state management system for task tracking.
- **DevOps:** Managed the full CI/CD pipeline for a 48-hour hackathon, ensuring 99.9% uptime during the competitive judging phase via Railway.

Link <https://tasklink-production.up.railway.app/landing>

LANGUAGE SKILLS

Mother tongue(s): **ROMÂNĂ**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLEZĂ	B2	B2	B2	B2	B2
GERMAN	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HOBBIES AND INTERESTS

Hobby-uri

Fitness, cooking and going on walks

VOLUNTEERING

29/09/2025 – CURRENT Timișoara

LigaAC member